

Feature Engineering and Conformal Prediction in Fish Classification.*

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Description of methods and main findings

1 Introduction

Hydroacoustic signals provide a non-lethal approach for fish classification. However, the features used for classification are high-dimensional and highly correlated. We are also often only given labels for a subset of the data, and must leverage both labelled and non-labelled data to improve classification accuracy. Lastly, we often care to understand which signals are most important for classification, but the correlation among features distorts this signal. The student will use various data science and machine learning approaches to address the data structure challenges and classification tasks.

2 Data

The data set contains **14,575 rows** and 484 columns. Each row represents one fish and it's measurement for a single "ping". A "ping" refers to a time the fish hydroacoustic signal was measured and stored in the dataset. Since multiple signals were recorded for a single fish, there are several rows corresponding the same fish (e.g. there are multiple rows for LT001; Lake Trout fish number one, because multiple measures were taken for this single fish). From now on, we call rows "pings", to familiarize ourselves with how the data was being collected.

The dataset contains information for three fish species: Lake Trout (LT), Lake Whitefish (LW) and Smallmouth Bass (SMB). Table 1 summarizes relevant information about the dataset for each of the species.

*Code and data are available at: <https://github.com/mejiama16/Fish-Project>. The data for this paper was retrieved from the [Fish Tether Experiment GitHub Repository](#).

Table 1: Overview of the dataset

Species	Number of fish	Total pings	Median pings	Min. pings	Max. pings
LT	21	7,248	213	16	1,379
LW	11	4,285	310	33	1,619
SMB	12	3,042	210	14	1,071

The data was obtained from 44 distinct fishes: 21 Lake Trout, 11 Lake Whitefish and 12 Smallmouth Bass. Lake Trout fishes has more pings (instances where data was collected from them), followed by Lake Whitefish and Smallmouth Bass. Although the LW species had less individual fish, more pings were obtained for this species than for SMB. The median pings column shows the median number of pings across all individual fishes for a certain species, in other words, what was the median number of instances where measurements were taken from a single fish. The values are fairly similar across LT and SMB, while LW had more pings in a single fish compared to the other. This is likely because the minimum and maximum number of pings obtained for LW is much higher than for LT and SMB.

It is important to understand the structure of the data and how it is collected, as it largely shapes the statistical method we select and how we make decisions throughout the process. In particular, the rows in this data do not represent single instances or observations, rather, each represents an instance of a subject in time; a measurement taken from a fish. As a result, it is possible for measurements obtained in these instances, although different, may be highly correlated as they come from the same fish. For instance, the rows 1-65 in the raw data set correspond to LT001, Lake Trout fish number one. Because all hydroacoustic signals are taken from this single fish, just at different points in time, it is possible for these to vary very little from measurement to measurement, resulting in highly correlated observations. Additionally, this feature of the data likely violates the assumption of independence of observations, that must hold from the simplest models (OLS regression) to the most complex models in machine learning. We are aware of this violation and consider its implications accordingly.

Now, one might think that the solution is to constraint our data set and analysis to just the 44 individual fish, by randomly selecting one of the measurements taken from each of them. The first issue with this approach is the resulting sample size. We have over 14,000 measurements obtained from this experiment, it seems like exploring the data to overcome the high correlation issue would be a much better solution, as compared to blindly throwing observations away. Another issue with this is: how do we pick which individual pings to keep and which to ignore? Do we hand-pick them? Do we select them randomly? Do we select two per day and ignore the rest? Lastly, when building machine learning models with this data, how can we ensure there is no data leakage? If we keep some observations from fish LT001 in the training set and some in the testing set, it is possible that due to the high correlation between these, the model has already memorized these patterns, leading to an overestimation of the model’s actual capacity.

This paper will touch on some of the decisions we must make when building models using this kind of data, and compare across them. We will do so in the context of dimensionality reduction and feature engineering. In particular, we want to get information regarding which features are the most informative in the classification task. That is, given all the signals we have available for a new, unknown fish, which of these signals are the most likely to help us decide whether this fish is a Lake Trout, a Lake Whitefish or a Smallmouth Bass.

2.1 Time Series Component

Before jumping into the methods it is important to better understand other characteristics of the data. First, the data contains a time series component. The pings for each fish are taken at different times throughout a period of time, no two pings happen simultaneously. Table 2 shows that the data was collected in the span of a week, with SMB having data collected in a much shorter period of time. Note that the data set included other dates, for example `processedDate`, however we keep `sampleDate` to reflect the time the measurement was taken.

Table 2: Span of the data collection process (pings)

Species	Earliest day a ping was made	Latest day a ping was made
LT	2022-07-21	2022-07-28
LW	2022-07-23	2022-07-28
SMB	2022-07-26	2022-07-28

As mentioned earlier, the data set contains a total of 484 columns, out of which 426 are hydro acoustic signals or frequencies. The frequencies go from F45 to F260, and they increase by 0.5. The other 58 columns contain information such as species, unique identifier for the fish and physical characteristics like sex, air bladder weight, tissue sample, length, among others. It is possible that these measures were taken via more aggressive methods; our analysis focuses on just the hydro acoustic signals, as a form to explore non-lethal methods to deal with species classification and problems adjacent to it.

For the purpose of this paper, we do not focus on the temporal data aspect of this problem, though it is possible to start training models based on this particular attribute of the data set. We leave this aspect to be explored in future research.

2.2 Missing Values

Table 3: Missing frequencies in the dataset

Species	F90-F170	F173-F260	Rows with any missing
LT	2,964	183	3,147
LW	789	0	789
SMB	306	0	306
Total	4,059	183	4,242

As observed in Table 3, our data set has 4,242 rows with missing values, most of which come from the Lake Trout species which is also the species for which the most data was collected. It is possible that some of these values are missing at random; for example, the part of the machine that gathered those frequencies was not working that day. While some imputation methods were considered to replace missing values, given the small number of fish in the sample, the consequences of imputing missing values are unknown, and specially when focusing on dimensionality reduction we want to avoid introducing more noise than there already is to perceive signals clearly. Consequently, we drop any rows that have missing values in the frequencies (4,242 in total).

After removing rows with missing frequencies, the resulting data set contains 10,333 rows. Table 4 shows the resulting data set. The number of LT fish is reduced by half. Based on these results, the initial dimensionality reduction will be performed in a binary classification setting between Lake Trout (LT) and Lake Whitefish (LW), given that they are the classes with the most pings (4,101 and 3,496), compared to Smallmouth Bass (2,736).

Table 4: Overview of the dataset with no missing values

Species	Number of fish	Total pings	Median pings	Min. pings	Max. pings
LT	12	4,101	189.5	16	1,379
LW	8	3,496	312.0	96	1,619
SMB	10	2,736	210.0	25	1,071

3 Methods

In the following section we attempt several techniques for variable selection (i.e. dimensionality reduction). We start with some regularization techniques, which fit a model containing all p predictors that constraints the coefficient estimates in such way that shrinks them towards zero. We attempt the common Lasso regularization and a new variation of Lasso, known as Whitening Lasso or WLasso, which serves in context high-dimensional, highly correlated data. We then explore more complex methods like PCA, random forest and boosting.

3.1 Lasso Regularization

Lasso regularization is a type of shrinkage regression that employs the l_1 penalty. This has the effect of forcing some coefficients to be exactly zero when the tuning parameter λ is sufficiently large. As a result of its construction, lasso returns a sparse model, which allows for variable selection and greater interpretability. Note that since we are focusing on a binary classification problem, we use the Lasso penalty with logistic regression instead of linear regression.

The first important consideration before fitting the model is how to split the data into test, train and validation. We need one set to train the initial model, one validation set to find the penalty λ that minimizes mean squared error and one test set to evaluate the performance of the final model, on new, unseen data. To pick the optimal λ , we use cross-validation. The main drawback is that aren't getting the best possible model in the first fitting round because we are not using all of the data, however this provides the best way to tune the parameter so we will proceed and evaluate overall model performance only using the test data.

It is clear that this split has to be done by fish, rather than by pings because otherwise our model would be exposed to a risk of data leakage. In other words, we may train the model on observations that are also in the test set (for example, if they come from the same fish), resulting in an overestimation of the model's performance. Doing the splits by fish ensures there is no data leakage introduced by our procedure, however it does limit us because we only have 20 fish in total after the data cleaning (12 LT and 8 LW). Adding to this, different fish do not have the same number of pings (total rows), as a result, and while some individual fish may have over 1000 pings, some may have as little as 16. If selecting via random sampling there is a possibility that the test set will end up with more total observations than the training set, which is not ideal as we always hope to train the model on the largest portion of data.

To overcome these issues, we attempt two methods: 1) train/validation/test split with stratified random sampling and 2) leave-one-fish-out. For the first method, we split the 20 fishes into 10 in the training set, 4 in the validation set and 6 in the testing set. For the second method, we use leave-one-out cross validation. This allows us to use most of the data in the training run and just one or two observations in the validation run. However, in this case we train the model multiple times and tune the parameter multiple times, and at the end we aggregate the results to obtain...

4 Results

5 Discussion

6 Appendix